Meaning of Frequency

Definition

Frequency is the number of times a particular value occurs in a dataset.

Illustration of Frequency Counting

Categories	Tally Marks	Frequency
Red	𝍸	5
Green	𝍸 |	6
Blue	||	2
Black	|	1
Yellow	|||	3

Groups	Tally	Frequency	Relative frequency
$0 < x \leq 10$	𝍸 |	6	
$10 < x \leq 20$	𝍸	5	
$20 < x \leq 30$	||||	4	
$30 < x \leq 40$	||	2	
$40 < x \leq 50$	𝍸	5	
$50 < x \leq 60$	||	2	
$60 < x \leq 70$		0	
$70 < x \leq 80$	|	1	
		Total =	

Example

Test scores:
15, 12, 15, 18, 12, 15, 20, 12, 18, 15

Score	Tally	Frequency
12		
15		
18		
20		
Total		**10**

What Frequency Tells Us

- Number of occurrences
- Most/least common values
- Pattern of distribution

Types of Data

1. Discrete Data

- Countable values
- Whole numbers only
  Examples:
- Number of students
- Goals scored
- Shoe sizes

2. Continuous Data

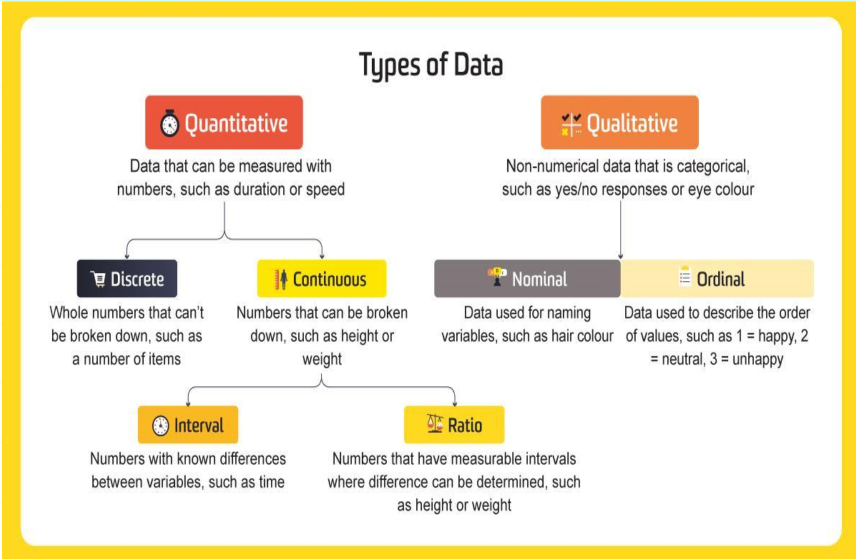
- Measurable values
- Can take any value in a range

Examples:

- Height
- Weight
- Time

Visual Comparison

Points	Discrete Data	Continuous Data	
Meaning	Discrete data has clear spaces between values.	Continuous data falls on a continuous sequence.	
Can you count the data?	Yes, data is usually units counted in whole numbers.	Generally, No	
Can you measure the data?	No	Yes	
Values	It has a infinite number of possible values. the values cannot be divided into smaller pieces and add additional meaning.	It has a infinite number of possible values. within an interval. the values can be subdivided into smaller and smaller pieces.	
Graphical Representation	Bar Chart	Histogram	
Examples	The number of students in a class The number of Workers in a Company.	The amount of time required to complete a project The height of Children.	



B. Construction of Frequency Tables

1. Ungrouped Data (Discrete Data)

Example

Number of children in 20 families:
2, 3, 4, 2, 5, 3, 2, 4, 3, 6, 3, 2, 4, 3, 5, 4, 3, 2, 4, 3

Steps

1. List values
2. Count frequencies

Number of Children	Tally	Frequency
2		
3		
4		
5		
6		
Total		**21**

Note: Total frequency = total observations

2. Grouped Data (Continuous Data)

Example

Heights of students grouped into intervals:

Height (cm)	Frequency
145–149	2
150–154	3
155–159	5
160–164	4
165–169	5
170–174	3
Total	**30**

Grouped Table Visualization

Complete Frequency Table

Table 2-6

Grouped Frequency Distribution for the Test Scores of 52 Students in Statistics

Class Intervals (*ci*)	Frequency (*f*)	Class Mark (*x*)	Class Boundary (*cb*)	Relative Frequency (*rf*)	<cf	>cf
0 - 2	20	1	-0.5 – 2.5	38.5%	20	52
3 – 5	14	4	2.5 – 5.5	26.9%	34	32
6 – 8	15	7	5.5 – 8.5	28.8%	49	18
9 – 11	2	10	8.5 – 11.5	3.8%	51	3
12 – 14	1	13	11.5 – 14.5	1.9%	52	1

Frequency Distribution

Frequency Distribution: A table that shows **classes** or **intervals** of data with a count of the number of entries in each class. It is used to organize data and helps to recognize patterns.

The **frequency, *f*,** of a class is the number of data entries in the class.

Class	Frequency
1 – 5	5
6 – 10	8
11 – 15	6
16 – 20	8
21 – 25	5
26 – 30	4

Larson/Farber 4th ed.

Key Terms in Frequency Distribution

1. Class Interval

Range of values (e.g., 20–29)

2. Class Width

Upper limit − Lower limit
Example: 29 − 20 = 9

3. Class Boundaries

True limits
Example:
20–29 → 19.5–29.5

4. Class Midpoint

(Lower + Upper)/2

5. Frequency

Number of observations in a class

Steps to Construct Grouped Frequency Table

Frequency distribution and Histogram of Grouped Data

Problem :

In a Statistics test, the marks obtained by 70 students in a class is given as follows :

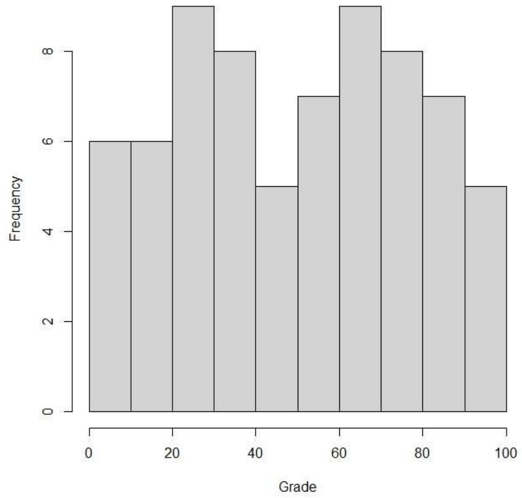
8 24 80 46 80 42 61 82 84 8 17 40 52 68 81
65 64 67 80 5 32 26 52 70 100 50 88 15 40 74
11 68 27 73 0 82 80 31 30 39 98 83 58 6 63
54 95 18 1 26 51 34 50 73 69 92 75 36 47 27
21 14 87 91 23 31 22 14 57 53

Prepare a frequency distribution of the above data.

Solution :

Frequency distribution :

Class Interval	frequency
0 - 10	6
10 - 20	6
20 - 30	9
30 - 40	8
40 - 50	5
50 - 60	7
60 - 70	9
70 - 80	8
80 - 90	7
90 - 100	5



Groups	Tally	Frequency	Cumulative Frequency
$20 < x \leq 30$	$\cancel{||||}$	5	5
$30 < x \leq 40$	$|||$	3	$5 + 3 = 8$
$40 < x \leq 50$	$\cancel{||||}$	5	$8 + 5 = 13$
$50 < x \leq 60$	$\cancel{||||}\ |||$	8	$13 + 8 = 21$
$60 < x \leq 70$	$||||$	4	$21 + 4 = 25$

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1. Find Range = Highest − Lowest
2. Choose number of classes (5–10)
3. Calculate class width
4. Create intervals
5. Tally data
6. Count frequency
7. Verify totals

C. Cumulative Frequency

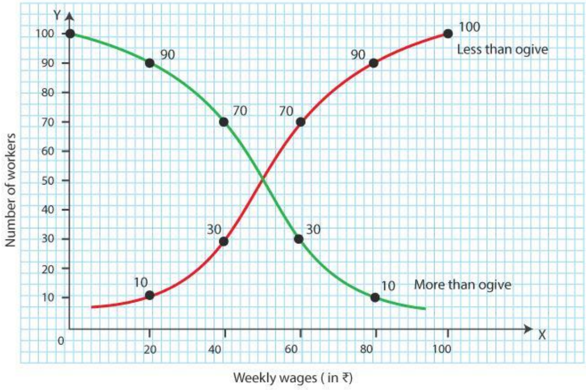
Definition

Cumulative frequency is the running total of frequencies.

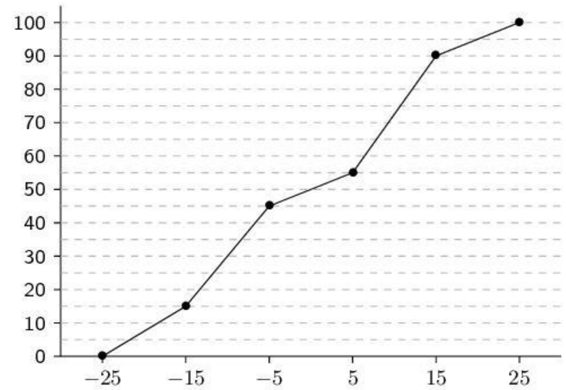
Example Table

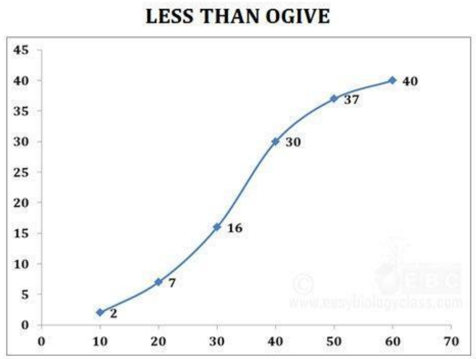
Class	Frequency	Cumulative Frequency
0–9	3	3
10–19	5	8
20–29	8	16
30–39	7	23
40–49	4	27
50–59	3	30

Cumulative Frequency Graph Idea



'Less than' and 'More than' Ogives





Types of Cumulative Frequency

1. Less Than

- Shows values below a boundary

2. More Than

- Shows values above a boundary

Worked Examples (Cleaned)

Example 1 (Grouped Table)

Class intervals: 25–29, 30–34, 35–39, 40–44

Students should tally and complete the table.

Example 2 (Cumulative Frequency)

Score	Frequency	Cumulative
1–10	3	3
11–20	7	10
21–30	12	22
31–40	10	32
41–50	8	40

Example 3 (Class Boundaries)

20–29 → 19.5 – 29.5

Example 4 (Midpoint)

35–44 → 39.5

Example 5 (Modal Class)

Highest frequency = Modal class

Class Activities

1. Collect ages of 20 students and form a table
2. Group raw data into 5 classes
3. Compute cumulative frequencies

Evaluation Questions

1. Define frequency
2. Difference between grouped and ungrouped data
3. Find class width of 30–39
4. Find class boundaries of 15–19
5. Complete cumulative frequencies:
   5, 13, __, __

Assignment

1. Construct grouped and cumulative tables from given data
2. Find:
    - Total frequency
    - Class boundaries
    - Midpoints
3. Build grouped table for heights
4. Explain key terms
5. Solve missing frequency problem